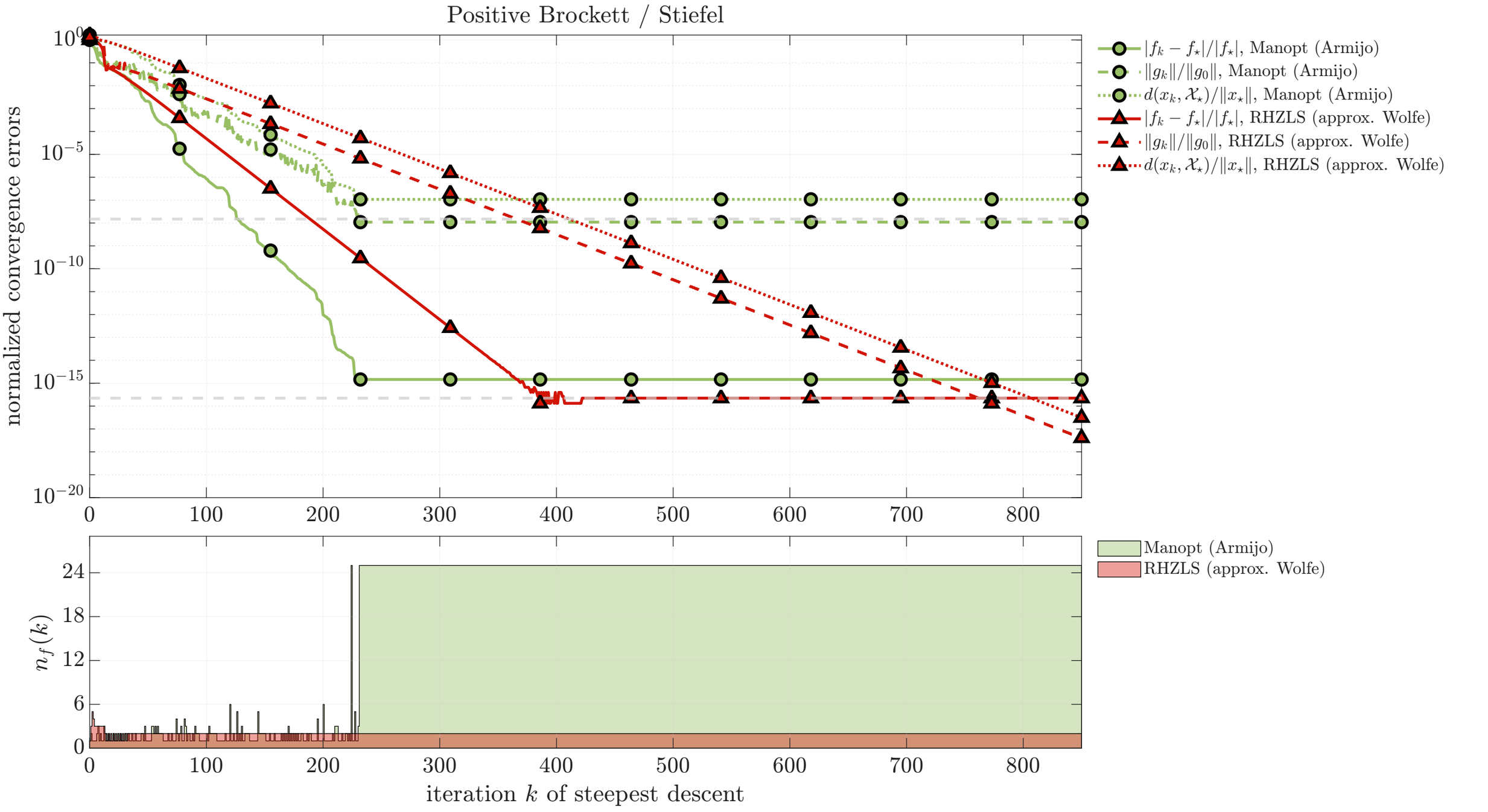

Positive Brockett / Stiefel
normalized convergence errors
10^{0}
10^{-5}
10^{-10}
10^{-15}
10^{-20}
0
100
200
300
400
500
600
700
800
$|f_k - f_\star|/|f_\star|$, Manopt (Armijo)
$\|g_k\|/\|g_0\|$, Manopt (Armijo)
$d(x_k, \mathcal{X}_\star)/\|x_\star\|$, Manopt (Armijo)
$|f_k - f_\star|/|f_\star|$, RHZLS (approx. Wolfe)
$\|g_k\|/\|g_0\|$, RHZLS (approx. Wolfe)
$d(x_k, \mathcal{X}_\star)/\|x_\star\|$, RHZLS (approx. Wolfe)
$n_f(k)$
24
18
12
6
0
Manopt (Armijo)
RHZLS (approx. Wolfe)
iteration k of steepest descent